

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

## Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

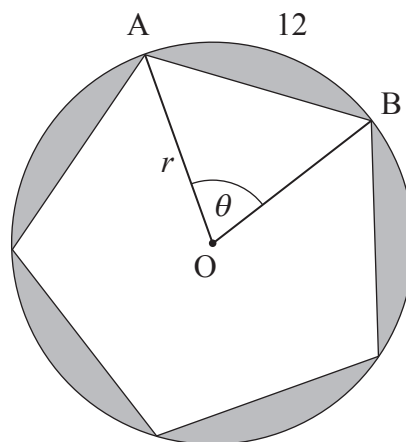
1. [Maximum mark: 6]

The following diagram shows a regular pentagon inscribed in a circle with centre  $O$  and radius  $r$  cm.

The angle  $\widehat{AOB}$  is  $\theta$ , where  $\theta$  is measured in radians.

The arc  $AB$  is 12 cm.

diagram not to scale



(a) Find

(i)  $\theta$ ;

(ii)  $r$ .

[3]

(b) Find the area of the shaded region.

[3]

(This question continues on the following page)



(Question 1 continued)

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Turn over